

1 Solutions of exercises in §1 (Policy Iteration)

Exercise 1.1 [*Policy Improvement Theorem*] Let π be any policy and let $\pi' = \text{Greedy}(V^\pi)$ - in other words, let $\pi'(s) = \operatorname{argmax}_a Q^\pi(s, a)$. Show that for every state s :

$$V^\pi(s) \stackrel{(1)}{\leq} Q^\pi(s, \pi'(s)) \stackrel{(2)}{\leq} V^{\pi'}(s)$$

Solution. We first prove inequality (1). Since π' is greedy with respect to V^π , it implies that $Q^\pi(s, \pi'(s)) \geq Q^\pi(s, \hat{\pi}(s))$ for any policy $\hat{\pi}$ and thus also for $\hat{\pi} = \pi$. Together with the fact that $V^\pi(s) = Q^\pi(s, \pi(s))$ we have:

$$V^\pi(s) = Q^\pi(s, \pi(s)) \leq Q^\pi(s, \pi'(s))$$

Next we illustrate two different approaches for showing (2):

First approach: This method (illustrated on page 78 in [1]) uses full expressions and ensures correctness but takes a bit more space. To prove (2) we can write:

$$\begin{aligned} Q^\pi(s, \pi'(s)) &= \mathbb{E}[r(s_0, a_0) + \gamma V^\pi(s_1) | s_0 = s, a_0 = \pi'(s)] \\ &= \mathbb{E}_{\pi'}[r(s_0, a_0) + \gamma V^\pi(s_1) | s_0 = s] \end{aligned}$$

[Note: The first expectation is taken over the next state, $s_1 \sim P(\cdot | s, \pi'(s))$. Since the policy π' is deterministic, conditioning on the chosen action does not introduce any additional randomness. Therefore, the second expectation is identical to the first one. In what follows, when we write $\mathbb{E}_{\pi'}$, we refer to the expectation over transition randomness, assuming that actions are selected according to the deterministic policy π' .]

Next, we can then apply inequality (1) to $V^\pi(s_1)$ to show $V^\pi(s_1) \leq Q^\pi(s_1, \pi'(s_1))$. By continuing this recursive reasoning step-by-step we get:

$$\begin{aligned} Q^\pi(s, \pi'(s)) &\leq \mathbb{E}_{\pi'}[r(s_0, a_0) + \gamma r(s_1, a_1) + \gamma^2 V^\pi(s_2) | s_0 = s] \\ &\vdots \\ &\leq \mathbb{E}_{\pi'}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s\right] = V^{\pi'}(s), \end{aligned}$$

Second approach: Here we use vector notation and matrix inversion rather than infinite recursion. First, note that $\mathcal{L}_{\pi'} V^\pi(s) = Q^\pi(s, \pi'(s))$. Thus, we can write inequality (1) equivalently as $V^\pi \leq \mathcal{L}_{\pi'} V^\pi$.

So, in order to prove (2) we are going to prove that $V^{\pi'} - \mathcal{L}_{\pi'} V^\pi \geq 0$. Now using the short notation, for any policy π we have $\mathcal{L}_\pi V = r_\pi + \gamma P_\pi V$. We can specialize this equation for $V = V^\pi$ and recall that $\mathcal{L}_\pi V^\pi = V^\pi$ to obtain $V^\pi = r_\pi + \gamma P_\pi V^\pi$, or equivalently $V^\pi = (I - \gamma P_\pi)^{-1} r_\pi$. Then, we have:

$$\begin{aligned} V^{\pi'} - \mathcal{L}_{\pi'} V^\pi &= (I - \gamma P_{\pi'})^{-1} r_{\pi'} - \mathcal{L}_{\pi'} V^\pi \\ &= (I - \gamma P_{\pi'})^{-1} (r_{\pi'} - (I - \gamma P_{\pi'}) \mathcal{L}_{\pi'} V^\pi) \\ &= (I - \gamma P_{\pi'})^{-1} \underbrace{(r_{\pi'} + \gamma P_{\pi'} \mathcal{L}_{\pi'} V^\pi - \mathcal{L}_{\pi'} V^\pi)}_{\mathcal{L}_{\pi'}(\mathcal{L}_{\pi'} V^\pi)} \\ &= (I - \gamma P_{\pi'})^{-1} \mathcal{L}_{\pi'} (\mathcal{L}_{\pi'} V^\pi - V^\pi) \end{aligned}$$

From (1) we have that $\mathcal{L}_{\pi'} V^\pi - V^\pi \geq 0$. Combining the facts that for $\gamma \in [0, 1]$ matrix $I - \gamma P_{\pi'}$ is positive definite and $\mathcal{L}_{\pi'}$ is always a non-negative operator (as rewards and transitions are always non-negative), we obtain that the right hand side is non-negative. Thus we have $V^{\pi'} - \mathcal{L}_{\pi'} V^\pi \geq 0$, as we wanted to show.

The two approaches are related by the von-Neumann series $(I - \gamma P_{\pi'})^{-1} = \sum_{t=0}^{\infty} \gamma^t P_{\pi'}^t$. In other words, we replaced infinite recursion from the first approach with matrix inversion in the second.

Exercise 1.2 [Convergence of PI] Show that the PI algorithm converges to an ε -optimal policy after at most $k = \log\left(\frac{1}{\varepsilon(1-\gamma)}\right) / \log\left(\frac{1}{\gamma}\right)$ iterations.

Solution. First note that by definition $V^* \geq V^\pi$ for any policy π . Thus we have that $V^* - V^{\pi_k} \geq 0$. Using inequality (2) from Exercise 1.1 we have that $-V^{\pi_k} \leq -\mathcal{L}V^{\pi_{k-1}}$ implying that $V^* - V^{\pi_k} \leq V^* - \mathcal{L}V^{\pi_{k-1}}$. Next, since both sides of the inequality are positive we can take absolute value and maximum over states to show that:

$$\|V^* - V^{\pi_k}\|_\infty \leq \|V^* - \mathcal{L}V^{\pi_{k-1}}\|_\infty$$

Next we use that $\mathcal{L}V^* = V^*$ and that Bellman operator is γ contraction to obtain:

$$\|V^* - \mathcal{L}V^{\pi_{k-1}}\|_\infty = \|\mathcal{L}V^* - \mathcal{L}V^{\pi_{k-1}}\|_\infty \leq \gamma \|V^* - V^{\pi_{k-1}}\|_\infty$$

Repeating this two steps for $k-1, k-2, \dots, 0$ we obtain:

$$\|V^* - V^{\pi_k}\|_\infty \leq \gamma \|V^* - V^{\pi_{k-1}}\|_\infty \leq \gamma^2 \|V^* - V^{\pi_{k-2}}\|_\infty \leq \dots \leq \gamma^k \|V^* - V^{\pi_0}\|_\infty$$

Finally we have $\|V^* - V^{\pi_0}\|_\infty \leq \|V^*\|_\infty \leq \frac{1}{1-\gamma}$ since all rewards $\in [0, 1]$. Thus:

$$\|V^* - V^{\pi_k}\|_\infty \leq \frac{\gamma^k}{1-\gamma} =: \varepsilon \implies k = \frac{\log\left(\frac{1}{\varepsilon(1-\gamma)}\right)}{\log\left(\frac{1}{\gamma}\right)}$$

where we took care of the fact that $\gamma \in (0, 1)$ when taking the logarithms. $\square$

2 Solutions of exercises in §2 (Supervised Learning)

Exercise 2.1 Determine π^* for this MDP:

$\pi^*(s_0) = \underline{\hspace{1cm}}, \pi^*(s_1) = \underline{\hspace{1cm}}, \pi^*(s_2) = \underline{\hspace{1cm}}$

Solution. Since the agent obtains reward only when taking action a_2 in state s_1 it is clear that optimal policy should lead to state s_1 from any other state and take action a_2 in s_1 :

$$\pi^*(s_0) = a_1, \pi^*(s_1) = a_2, \pi^*(s_2) = a_1$$

Exercise 2.2 Recall that s_0 is the fixed initial state. Determine d^* :

$d^*(s_0) = \underline{\hspace{1cm}}, d^*(s_1) = \underline{\hspace{1cm}}, d^*(s_2) = \underline{\hspace{1cm}}$.

Solution. Note that s_0 is always initial state and also transient state since the agent cannot come back to it after first time step. Thus we have $d^*(s_0) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t^\pi = s_0) = (1-\gamma) \gamma^0 \mathbb{P}(s_0^\pi = s_0) = 1-\gamma$. Since $\pi^*(s_0) = a_1$ and $\pi^*(s_1) = a_2$ this means that following policy π^* we never visit state s_2 and just stay in state s_1 for time steps $t = 1, 2, \dots$. Thus $d^*(s_1) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t^\pi = s_0) = (1-\gamma) \sum_{t=1}^{\infty} \gamma^t \mathbb{P}(s_t^\pi = s_1) =$

$(1 - \gamma) \frac{\gamma}{1 - \gamma} = \gamma$. So the solution is:

$$d^* = (1 - \gamma, \gamma, 0)$$

Exercise 2.3 Show that policy $\hat{\pi}$ fits the expert data almost perfectly:

$$\mathbb{P}_{s \sim d^*}(\hat{\pi}(s) \neq \pi^*(s)) = \varepsilon$$

Solution. From solution of Exercise 2.1 and definition of $\hat{\pi}$ we see that

$$\mathbb{P}(\hat{\pi}(s) \neq \pi^*(s) | s) = \begin{cases} \frac{\varepsilon}{1 - \gamma}, & \text{if } s = s_0 \\ 0, & \text{if } s = s_1 \\ 1, & \text{if } s = s_2 \end{cases}$$

Using conditional probabilities we have:

$$\begin{aligned} \mathbb{P}_{s \sim d^*}(\hat{\pi}(s) \neq \pi^*(s)) &= \sum_{k=0,1,2} \mathbb{P}(\hat{\pi}(s) \neq \pi^*(s) | s = s_k) \mathbb{P}_{s \sim d^*}(s = s_k) \\ &= \frac{\varepsilon}{1 - \gamma} \cdot (1 - \gamma) + 0 \cdot \gamma + 1 \cdot 0 = \varepsilon \end{aligned}$$

Exercise 2.4 First, calculate $d^{\hat{\pi}}$: $d^{\hat{\pi}}(s_0) = \underline{\hspace{1cm}}$, $d^{\hat{\pi}}(s_1) = \underline{\hspace{1cm}}$, $d^{\hat{\pi}}(s_2) = \underline{\hspace{1cm}}$. Then, show that under policy $\hat{\pi}$, the error is amplified as follows:

$$\mathbb{P}_{s \sim d^{\hat{\pi}}}(\hat{\pi}(s) \neq \pi^*(s)) = \frac{\varepsilon}{1 - \gamma}.$$

Solution. Here we explain how to determine $d^{\hat{\pi}}(s_1)$ and the other two values can be deduced similarly. Note that by following policy $\hat{\pi}$ the only path that leads us to the state s_1 is taking action a_1 in s_0 . So we have:

$$\begin{aligned} d^{\hat{\pi}}(s_1) &= (1 - \gamma) \sum_{t=1}^{\infty} \gamma^t \mathbb{P}(s_t^{\pi} = s_1) = (1 - \gamma) \mathbb{P}(a_0^{\pi} = a_1) \sum_{t=1}^{\infty} \gamma^t \\ &= (1 - \gamma) \left(1 - \frac{\varepsilon}{1 - \gamma}\right) \frac{\gamma}{1 - \gamma} = \gamma \left(1 - \frac{\varepsilon}{1 - \gamma}\right). \end{aligned}$$

The final solution is given by:

$$d^{\hat{\pi}} = \left(1 - \gamma, \left(1 - \frac{\varepsilon}{1 - \gamma}\right) \gamma, \frac{\gamma \varepsilon}{1 - \gamma}\right)$$

The second part of the exercise can be solved similarly to what we did in Exercise 2.3. We have:

$$\begin{aligned} \mathbb{P}_{s \sim d^{\hat{\pi}}}(\hat{\pi}(s) \neq \pi^*(s)) &= \sum_{k=0,1,2} \mathbb{P}(\hat{\pi}(s) \neq \pi^*(s) | s = s_k) \mathbb{P}_{s \sim d^{\hat{\pi}}}(s = s_k) \\ &= \frac{\varepsilon}{1 - \gamma} (1 - \gamma) + 0 \cdot \gamma \left(1 - \frac{\varepsilon}{1 - \gamma}\right) + 1 \cdot \frac{\gamma \varepsilon}{1 - \gamma} = \frac{\varepsilon}{1 - \gamma} \end{aligned}$$